ModelIt!

Teacher's Guide

Metabolic Engineering for Biofuels

Pathway Optimization and Flux Balancing in Microbial Fuel Production

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS1-5, HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-5, HS-ETS1-3

Use a model to illustrate how photosynthesis transforms light energy into stored chemical energy; optimize the design of a solution to a complex real-world problem by considering trade-offs among criteria and constraints.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Using Mathematics and Computational Thinking** — Students develop a metabolic flux model integrating enzyme kinetics, cofactor balance, and growth-production trade-offs, using quantitative analysis to optimize pathway performance.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow / ETS1.C Optimizing the Design Solution** — Matter and energy flow through metabolic pathways following conservation laws; engineering design requires iterative optimization of multiple competing criteria and constraints.
Crosscutting Concept	**Energy and Matter / Stability and Change** — Students trace carbon and energy flow from substrate through metabolic pathways to product, analyzing how perturbations in one pathway component cascade through the system to affect overall stability and output.

CAST Testing Alignment

MULTIPLE CHOICE:

An engineered E. coli strain producing butanol shows high Enzyme Activity but Fermentation Efficiency is declining over a 48-hour fermentation. Based on your model, which factor is most likely responsible?

CONSTRUCTED RESPONSE:

Using your metabolic engineering model, explain why doubling the expression of every enzyme in a biofuel pathway does NOT double the Product Yield. Reference at least three model components and describe the trade-offs involved.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to metabolic engineering — the systematic design and optimization of cellular metabolic pathways for industrial production. Biotech skill focus: Pathway flux balancing and multi-variable optimization. The global energy crisis demands alternatives to fossil fuels, and engineered microorganisms represent one of the most promising routes to sustainable liquid fuels.

The Energy Challenge and Biofuels

Global civilization consumes approximately 100 million barrels of crude oil per day — roughly 15.9 billion liters. Transportation fuels (gasoline, diesel, jet fuel) account for about 60% of this consumption. First-generation biofuels (corn ethanol, biodiesel from vegetable oil) have demonstrated the concept of biological fuel production but face fundamental limitations: they compete with food production for agricultural land, their energy return on investment (EROI) is marginal (1.3:1 for corn ethanol versus 15:1 for conventional oil), and ethanol has only 67% the energy density of gasoline. Advanced biofuels — produced by engineered microorganisms from non-food feedstocks like agricultural waste, forestry residues, or CO2 — could overcome these limitations if metabolic engineering can close the gap between laboratory yields and commercial requirements.

Metabolic Pathway Design

Engineering a microbial biofuel pathway involves three fundamental challenges. First, pathway selection: choosing the biochemical route from substrate to product. For example, fatty acid-derived biodiesel requires a pathway from glucose through acetyl-CoA to fatty acid synthesis to ester formation — a 12+ enzyme cascade. Butanol production through the Clostridial ABE pathway requires 7 enzymes. Each route has a different theoretical maximum yield set by biochemical stoichiometry and thermodynamics. Second, enzyme balancing: each enzyme in the pathway must be expressed at the right level. Too little of the bottleneck enzyme limits flux; too much of any enzyme wastes cellular resources. Third, cofactor engineering: most biosynthetic pathways consume reducing equivalents (NADH or NADPH) and ATP. If the pathway drains these cofactors faster than central metabolism regenerates them, flux stalls regardless of enzyme levels.

The Metabolic Burden Problem

Every protein a cell makes costs resources: ribosomes to translate the mRNA, amino acids to build the protein, ATP to fold it, and chaperones to maintain it. When metabolic engineers introduce 5-15 foreign enzymes at high expression levels, the total protein production burden can consume 20-40% of the cell's biosynthetic capacity. This 'metabolic burden' manifests as slower growth, reduced stress tolerance, and paradoxically lower product yield because the cell cannot maintain healthy metabolism while running the production pathway at full capacity. The solution is dynamic pathway regulation — genetic circuits that sense the cell's metabolic state and adjust enzyme expression in real time, similar to how natural metabolic regulation works. Strategies include growth-coupled production (linking product formation to essential metabolic pathways so cells cannot grow without producing), quorum sensing-based induction (delaying pathway activation until cell density is high), and stress-responsive promoters.

From Laboratory to Industrial Scale

The gap between laboratory biofuel production and commercial viability is measured in three metrics: yield (grams of product per gram of substrate — must be above 85% of theoretical maximum), titer (final product concentration — must exceed 50 g/L for economical downstream processing), and productivity (grams per liter per hour — determines bioreactor capital costs). Most laboratory strains achieve 30-60% of theoretical yield at titers of 10-20 g/L, well below commercial requirements. Scale-up introduces additional challenges: oxygen transfer limitations in large bioreactors (which can be 200,000+ liters), contamination risks during multi-day fermentations, the cost of sterilizing and cooling large volumes, and genetic stability over

hundreds of generations. Companies like Amyris, LanzaTech, and Genomatica have spent over a decade and billions of dollars bridging this gap for specific products.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine components spanning substrate input (Substrate Concentration), enzyme-level function (Enzyme Activity), pathway-level dynamics (Pathway Flux, Cofactor Availability), cell-level constraints (Metabolic Burden, Cell Growth Rate), and process-level outcomes (Product Yield, Byproduct Accumulation, Fermentation Efficiency).

E **Establish Relationships**

Students determine that Substrate Concentration is the primary external input, while Enzyme Activity, Pathway Flux, Cofactor Availability, Metabolic Burden, Cell Growth Rate, Product Yield, Byproduct Accumulation, and Fermentation Efficiency emerge as interconnected internal responses — recognizing that Enzyme Activity is partially controllable through genetic engineering but ultimately constrained by cellular resources.

V **Visualize & Model**

Students build a metabolic flux model connecting substrate input through enzymatic conversion steps to product output, visualizing how carbon flows partition between desired product, biomass, and unwanted byproducts.

E **Evaluate Outcomes**

Students run baseline, maximum-expression, and balanced-optimization scenarios to discover that maximum enzyme expression does not produce maximum Fermentation Efficiency due to Metabolic Burden effects.

R **Revise & Extend**

Students add Product Toxicity Threshold, Feedstock Flexibility, or Genetic Stability Index to explore more realistic industrial constraints on microbial biofuel production.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"We burn 100 million barrels of oil every day. What if, instead of drilling it out of the ground, we could brew it in a vat — using engineered microbes that convert sugar into fuel? That's metabolic engineering."*

Watch For

Display the scale comparison: barrels of oil consumed daily versus current biofuel production. Let the gap sink in.

Slide 2: Learning Objectives

3 min

Say: *"Today you're metabolic engineers. You're going to design a microbial fuel factory — and discover why the biggest challenge isn't the chemistry, it's the biology."*

Watch For

Pre-teach metabolic flux and cofactor recycling. Use the analogy: a factory assembly line where every worker needs specific tools (cofactors) to do their job.

Slide 3: Big Question

3 min

Say: *"Yeast has been converting sugar to ethanol for 10,000 years — that's beer and wine. But ethanol is a lousy fuel. What if we could redirect that metabolism to make jet fuel, diesel, or gasoline — drop-in replacements?"*

Watch For

Think-pair-share: Why is ethanol a poor fuel compared to gasoline? (Energy density, water miscibility, corrosion.) What properties would an ideal biofuel have?

Slide 4: LEVER Framework

2 min

Say: *"We're going to trace carbon from a sugar molecule all the way through an engineered metabolic pathway to a fuel molecule. Every step has constraints, trade-offs, and potential failure points."*

Watch For

Preview LEVER steps. Emphasize: this model captures the fundamental tension between making the cell a factory and keeping the cell alive.

Slide 5: Activity 1: Components

8 min

Say: *"Only ONE of these nine components is something the engineer fully controls from outside. The rest are biological responses. Which one?"*

Watch For

Guide identification of Substrate Concentration as the primary external input. Discuss how Enzyme Activity is partially controllable through genetic engineering but constrained by Metabolic Burden.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the problem: every carbon atom that enters the cell has three possible fates — it becomes fuel (what you want), biomass (what the cell wants), or byproduct (what neither of you wants). Trace the competition."*

Watch For

Students map carbon flow from Substrate through Pathway Flux to the three-way split: Product Yield, Cell Growth Rate (biomass), and Byproduct Accumulation. Identify Cofactor Availability as the hidden constraint.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run scenario 2 first — maximum enzyme expression. Watch what happens. Then figure out why 'more enzyme' made things WORSE. Then design the fix."*

Watch For

Students predict, then simulate. The counter-intuitive result — maximum expression crashes growth and total production — drives the key insight about Metabolic Burden. Challenge them to find the optimal balance.

Slide 8: Discoveries

5 min

Say: *"You just discovered the first law of metabolic engineering: the cell is not a beaker. It's a living system with its own agenda, and you have to negotiate with it, not dictate to it."*

Watch For

Connect findings to real industrial data. Show the gap between lab yields and commercial targets. Discuss what it takes to close that gap.

Slide 9: STEM Challenge

12 min

Say: *"A startup just gave your team $50 million. Design the strain, the pathway, and the fermentation process that makes jet fuel at $0.60 per liter. Can you compete with petroleum?"*

Watch For

Groups design complete biofuel production systems: host organism selection, pathway engineering strategy, fermentation process, and economic analysis. Present and defend designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Substrate Concentration
- **INTERNAL:** Enzyme Activity, Pathway Flux, Cofactor Availability, Metabolic Burden, Cell Growth Rate, Product Yield, Byproduct Accumulation, Fermentation Efficiency

Relationships

- **Substrate Concentration to Pathway Flux:** POSITIVE (+) — Higher substrate concentration provides more carbon molecules to enter the engineered pathway, increasing flux through the first enzymatic step up to the enzyme's saturation point.
- **Enzyme Activity to Pathway Flux:** POSITIVE (+) — More active pathway enzymes process intermediates faster, increasing the overall flux of carbon through the pathway toward the desired biofuel product.
- **Enzyme Activity to Metabolic Burden:** POSITIVE (+) — Higher enzyme expression diverts more cellular resources (ribosomes, amino acids, ATP) from growth to foreign protein production, increasing the fitness cost on the cell.
- **Metabolic Burden to Cell Growth Rate:** NEGATIVE (-) — Greater metabolic burden slows cell division by depleting resources needed for DNA replication, membrane synthesis, and essential protein production.
- **Pathway Flux to Cofactor Availability:** NEGATIVE (-) — Faster pathway flux consumes cofactors (NADH, NADPH, ATP) more rapidly, depleting intracellular pools and potentially stalling the pathway and other essential cellular processes.
- **Pathway Flux to Product Yield:** POSITIVE (+) — Higher pathway flux directs more carbon atoms toward the desired biofuel product rather than allowing them to be diverted to competing metabolic routes or byproducts.

Key Discoveries

- Maximum enzyme expression does NOT produce maximum product yield — excessive Metabolic Burden crashes Cell Growth Rate, resulting in fewer cells and lower total productivity than a moderately-expressed pathway
- Cofactor imbalance is a silent killer of engineered pathways — a pathway that looks perfect on paper fails in practice because it consumes NADH faster than the cell can regenerate it from central metabolism
- Byproduct accumulation is both a carbon sink and a toxin — acetate and organic acid byproducts divert 15-40% of substrate carbon while simultaneously inhibiting cell growth at concentrations above 5-10 g/L
- The economic breakeven for microbial biofuel production requires achieving at least 85% of theoretical maximum yield at titers above 50 g/L — most laboratory strains achieve 30-60% of maximum yield at 10-20 g/L, highlighting the engineering gap

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"More enzyme expression always means more product"	This is the most common and most dangerous misconception in metabolic engineering. The model clearly demonstrates that increasing Enzyme Activity beyond a critical threshold DECREASES total product output because Metabolic Burden crashes Cell Growth Rate. Fewer, sicker cells produce less total product than a moderate number of healthy cells expressing enzymes at balanced levels. The relationship between enzyme expression and product yield is a curve with a peak — and the peak is almost never at maximum expression.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Biofuels are automatically carbon-neutral because they come from plants"	The carbon accounting for biofuels is far more complex than 'plants absorb CO2, we burn biofuel, CO2 goes back to plants.' A full life cycle analysis must include: fossil energy used to grow and harvest the feedstock (tractors, fertilizer from natural gas, irrigation), energy to transport biomass to the biorefinery, energy to pretreat and sterilize the feedstock, energy for fermentation (aeration, cooling, mixing), energy for product separation and purification, and land use change emissions (clearing forest for agriculture releases stored carbon). When all these are counted, corn ethanol reduces greenhouse gas emissions by only 20-40% compared to gasoline — far from carbon-neutral.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We can just scale up the lab strain and it will work in a big fermenter"	Scale-up from a laboratory flask (milliliters) to an industrial bioreactor (hundreds of thousands of liters) changes almost everything about the microbial environment. Oxygen transfer becomes limited because bubbles can't reach all cells in a thick broth. Temperature gradients develop because heat from metabolism can't be removed fast enough. Mixing creates shear forces that damage cells. pH varies across the vessel. Contamination risk increases with longer fermentation times. And genetic instability means the production pathway mutates and degrades over hundreds of generations. The model's Fermentation Efficiency captures these scale-up losses, which typically reduce performance by 30-50% compared to laboratory results.